



Biochip Course

Introduction

Biosensors and Lab-on-a-Chip Systems

Marco Carminati

a.y. 2022/2023

Feb. 20th 2023



Course Info

5 CFU, 50 hours:

- 30h (15 days) theoretical lessons
- 12h (6 days) exercises, applicative lessons
- 4h (2 days) design lab (class discussion in teams)
- 4h (2 days) experimental lab (hands on PCB, fluidics and instrumentation)

Data	Dove	09:00	10:00	11:00	12:00	13:00	14:00	15:00	16:00	17:00	18:00
Lunedì	25.S.2										
Martedì											
Mercoledì	26.16										

BIOCHIP lezione (dal 20/02/2023 al 29/05/2023)

BIOCHIP lezione (dal 22/02/2023 al 31/05/2023)

No breaks, start at 16:30 sharp, end at 18:00



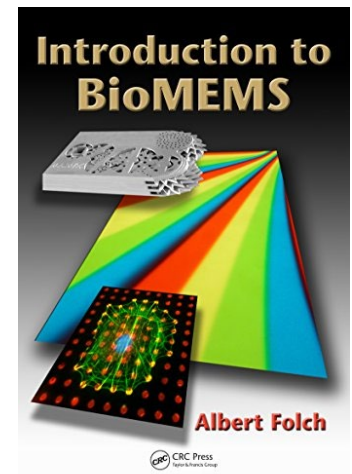
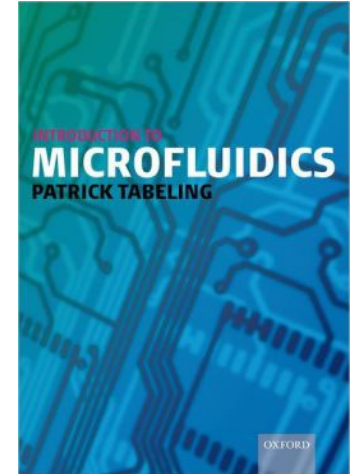
Background knowledge:

- **Physics:**
 - Electric charge, Coulomb force
 - Conductor, insulator, Ohm's law
- **Electronics:**
 - Operational amplifier
 - Feedback



Study Material:

- **Slides (on WeBeeP)** of lessons and exercises
- Webex lessons recordings
- Additional video material
- Past exams
- **Recommended textbooks:**
 - Patrick Tabeling, [Introduction to Microfluidics](#)
Oxford University Press, 2010
 - Albert Folch, [Introduction to BioMEMS](#)
CRC Press, 2016
- Additional material: papers, tutorials





Features

- **Focus examples on DNA and Corona Virus**
- **Hands-On Laboratory (May 29th and 31st)**
 - Teams of ~4 students, design of PCB @ home
 - Class discussion, fabrication and characterization in lab
- **Talks from Industry:**
 - STMicroelectronics (DNA)
 - others...
- **MS Forms/Socrative App** for self-evaluation quizzes
- **Additional videos** on fabrication (PoliFab cleanroom)





Marco Carminati
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Marco Bianchessi
STMicroelectronics



How to Contact Me

Please contact me by email:

marco1.carminati@polimi.it

in order to schedule a meeting (in person or via Teams)

Office location:



Ed. 24

Via Golgi, 40

Phone:

02.2399.6102



Webex Meetings

- Webex Meetings to attend lessons remotely and view recordings (links available on BeeP)

log-in with PoliMi credentials

<https://politecnicomilano.webex.com/meet/marco1.carminati>



WeBeep

- WeBeep: teaching material



- MS Teams: for one-to-one meetings



- MS Forms: self-evaluation tests



Exam Rules

Written Test (2hours)

2 dates in June/July

1 date in September

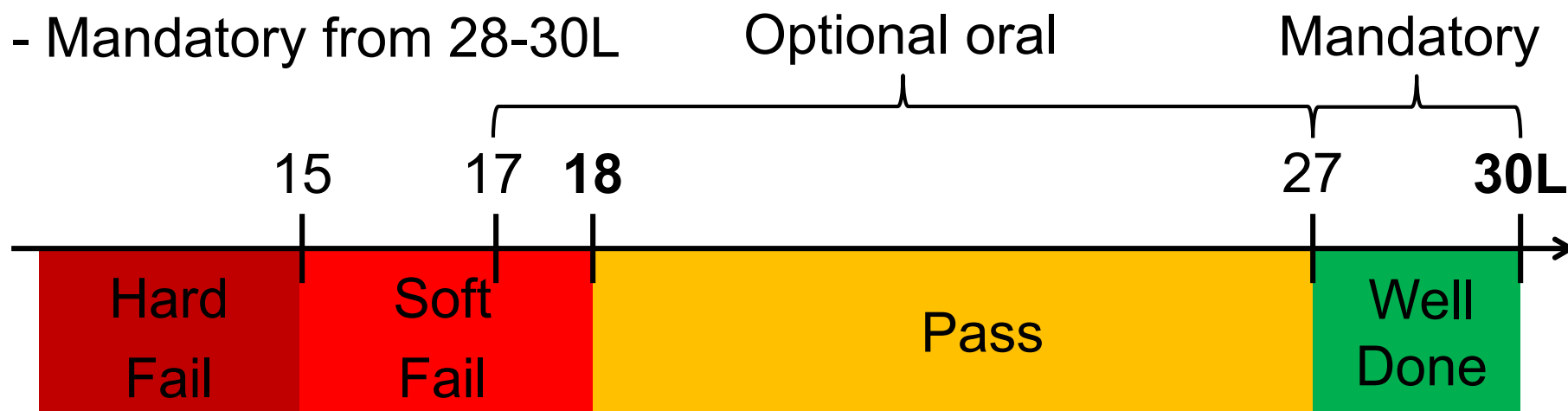
2 dates in Jan/Feb.

Oral Test:

- On my request
- Optional from 17 to 27
- Mandatory from 28-30L



Ritratto di Luca Pacioli





Biosensors: Two Meanings

Biosensor



A (chemical) sensor based on a
biological entity
(strong definition)



A sensor that monitors the status of
a biological system
(weak definition)



The Oldest Biosensors

The mine Canary: the first gas biosensor (full-animal)

Biological entities usable for sensing:

- Cell colony (tissue)
- Single Cell
- **Macromolecules**
- Enzyme
- Antibodies
- Nucleic acids

Affinity



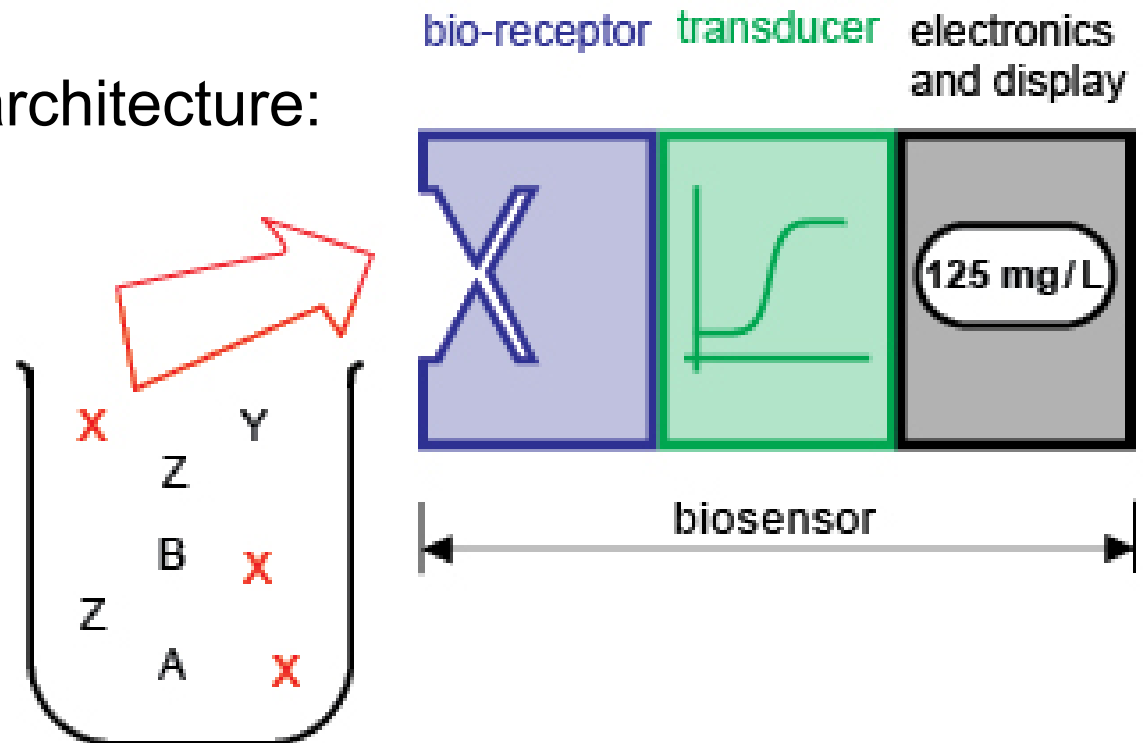


Analytical Biosensors

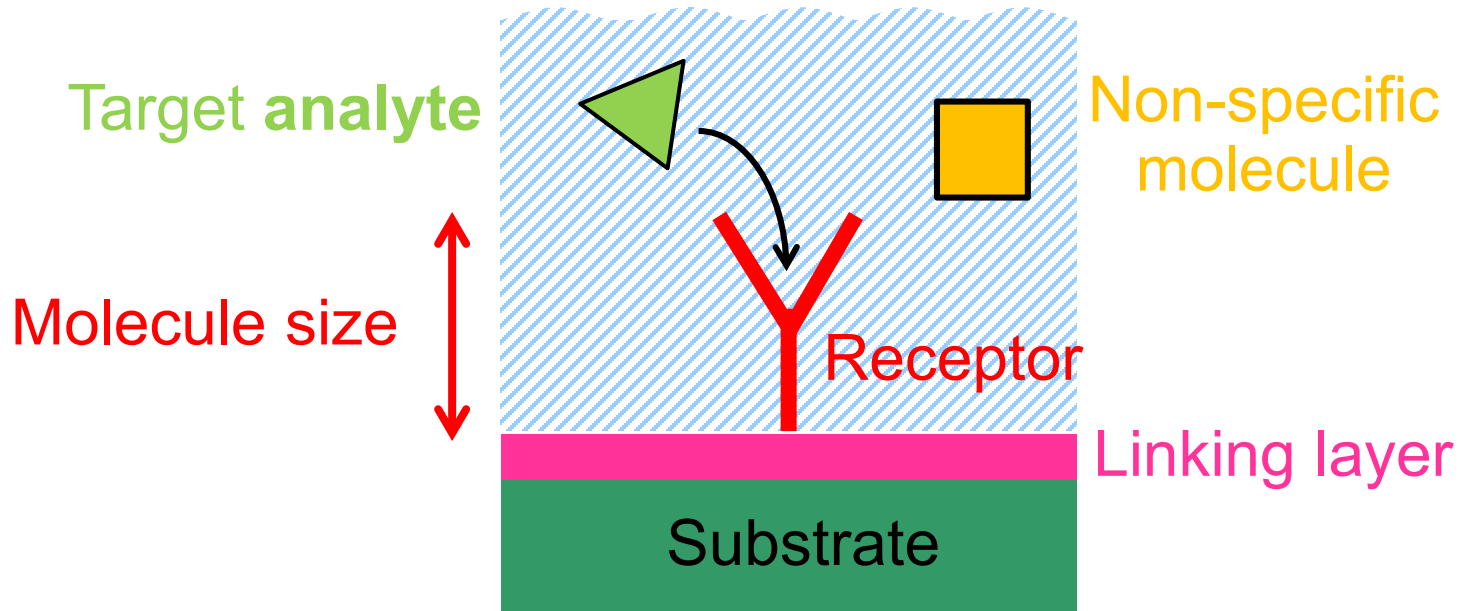
The **selectivity** (i.e. specificity only to analyte **x**) and the **sensitivity** of biological **receptors** are leveraged.

After the binding with the target analyte, some physical properties of the receptor change and are transduced into electrical signals.

Analytical biosensor architecture:



Planar configuration: the receptor is immobilized on the sensor



Technological issue: how to attach the molecule to a solid substrate while preserving its function (in a liquid environment)

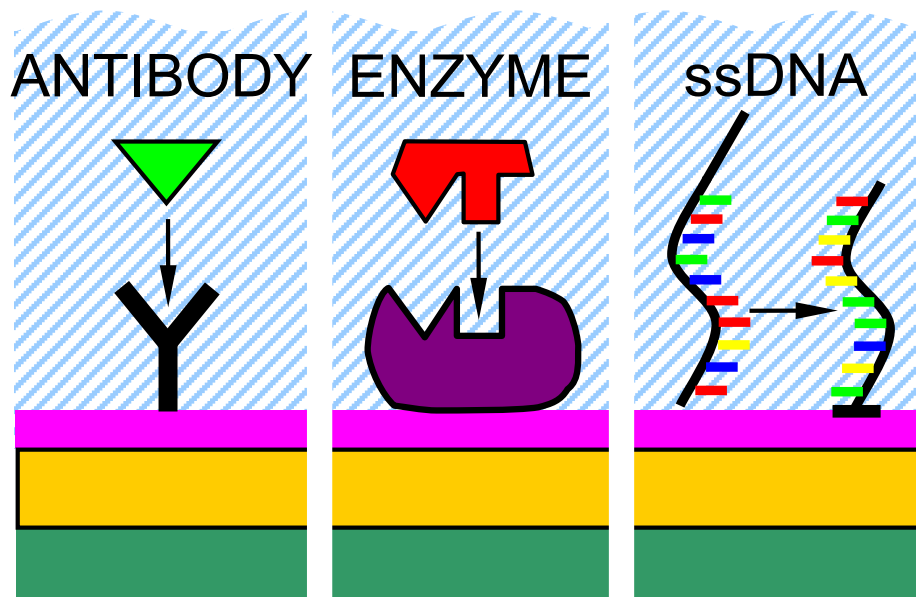
chemical functionalization



Nature's Catalog of Matching Molecules

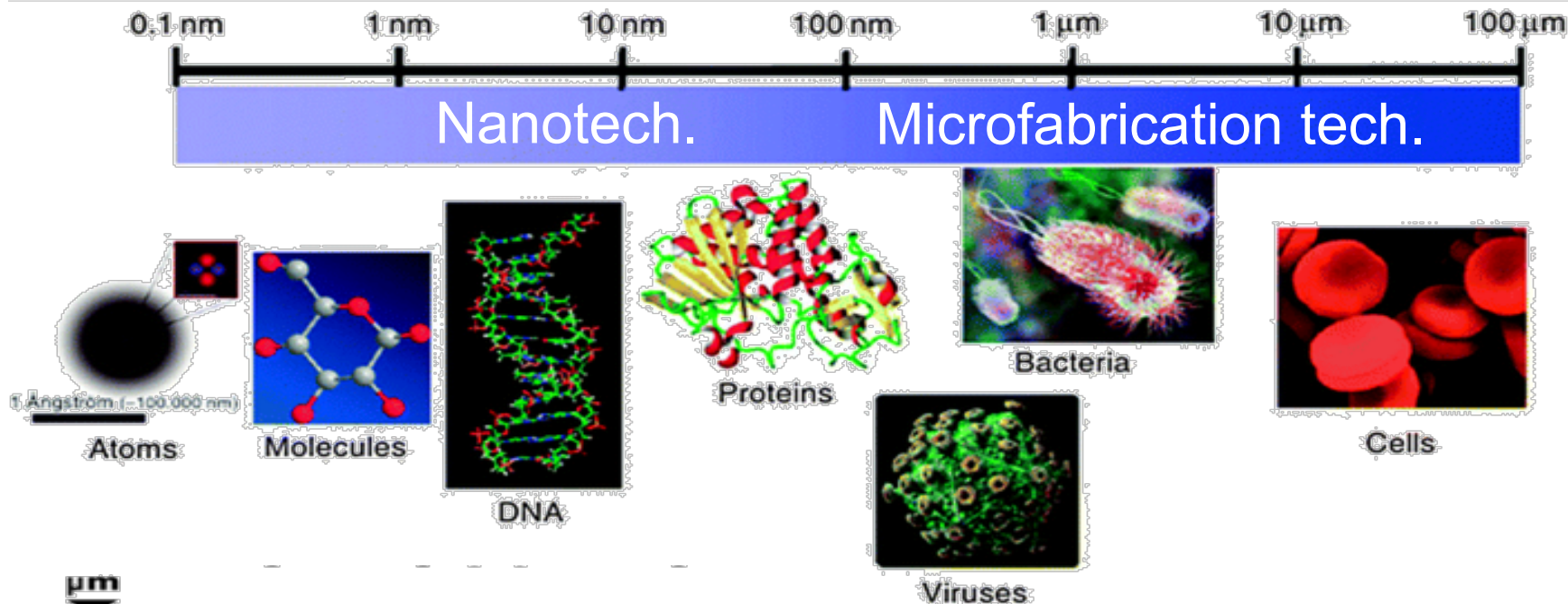
Several **molecular affinity** mechanisms are deployable:

- *Antibody-antigen* (immunosensors)
- *Enzyme-substrate* (enzymatic sensors)
- *DNA and nucleic acids* (genosensors)

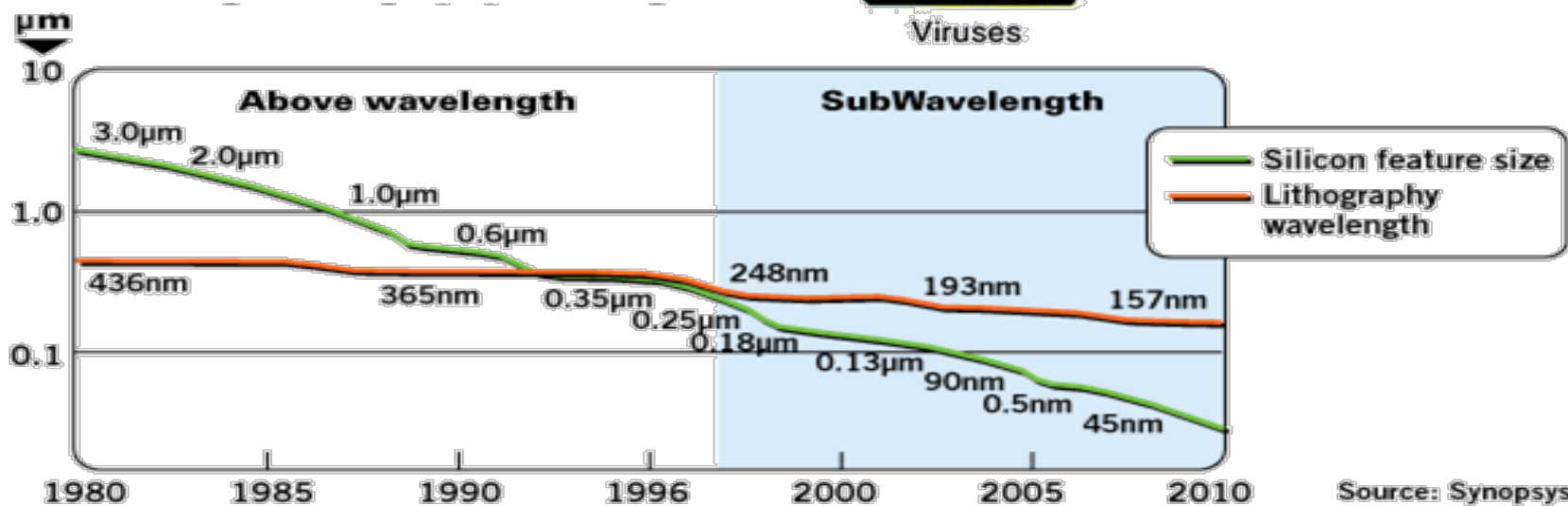


Size Scaling and Compatibility

Nature



Microelectronics





Size of Biological Entities: from Nano to Micro

Element	Size
Amino acid	0.8 nm
Diameter of a DNA helix	2 nm
Globular protein	4 nm
Thickness of a cellular membrane	10 nm
Diameter of a microtubule	25 nm
Large virus	100 nm
Lysosome	200 nm
Prokaryote	1–10 μm
Bacteria <i>E. Coli</i>	2 μm
Mitochondria	3 μm
Eukaryote	10–30 μm
Amoeba	90 μm
Frog oocyte	2 mm

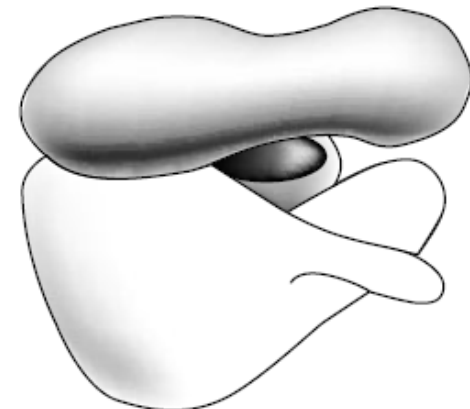
Sugar, Amino-acids,
Nucleotides ~ 0.5–1 nm



Globular proteins ~ 2–10 nm



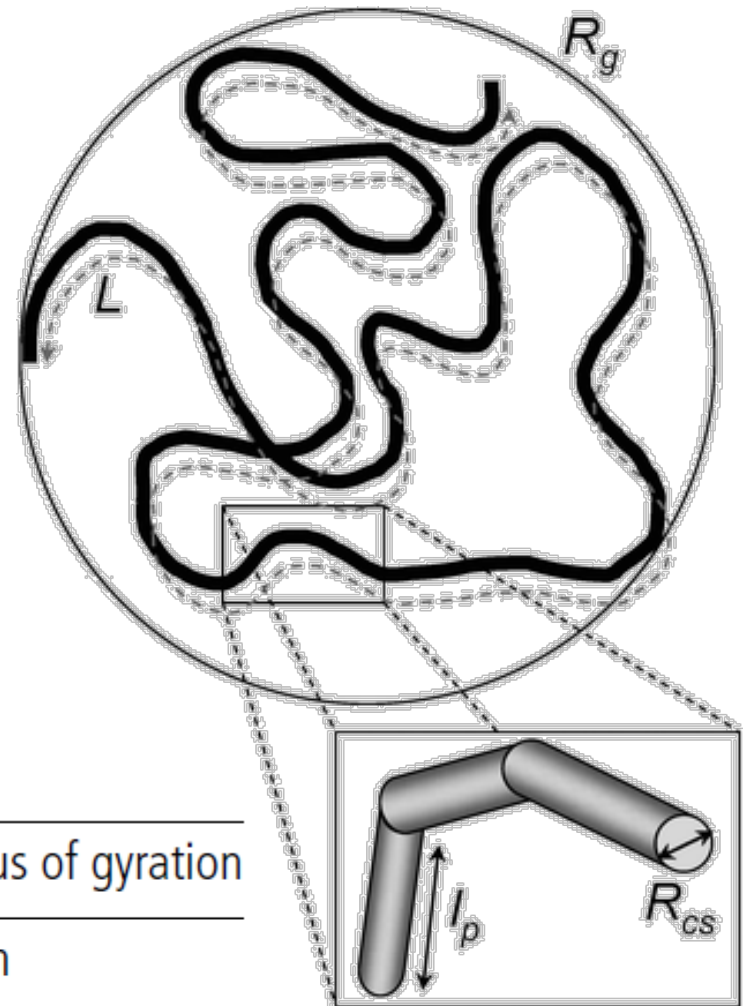
Ribosome ~ 30 nm





Geometrical Parameters

- **Contour length** (L): length of the macromolecule backbone (for a polymer chain is the maximum stretched length)
- **Radius of gyration** (R_g): for a folded, globular macromolecule is the average distance between the extremes

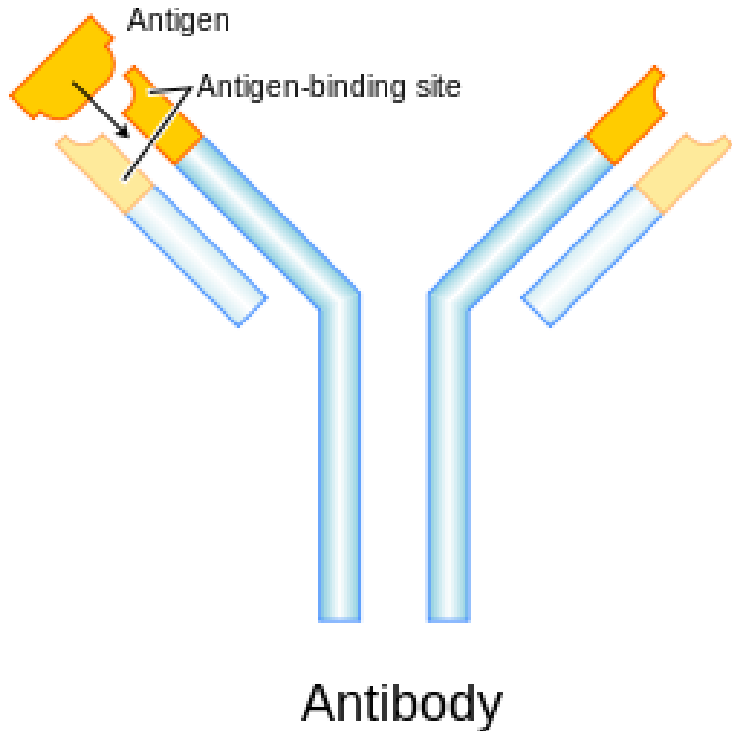
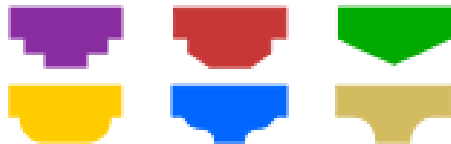


Molecule	Contour length	Radius of gyration
DNA of a human chromosome	5 cm	1 μm
DNA of the bacteria <i>E. Coli</i>	1.5 mm	100 nm



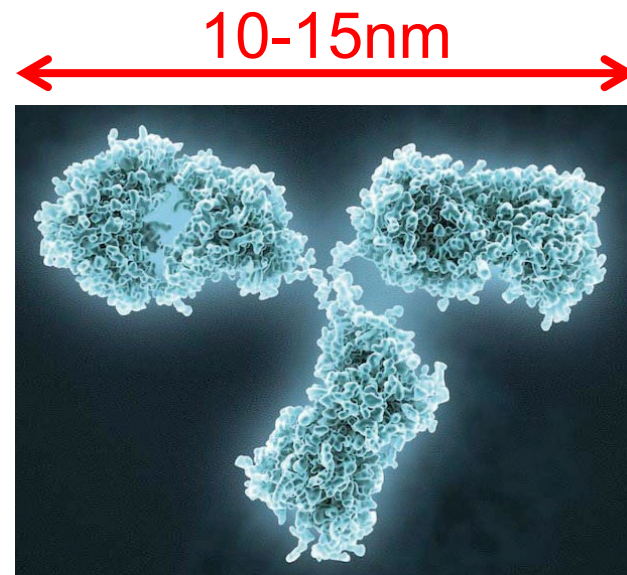
Antibodies

Antigens

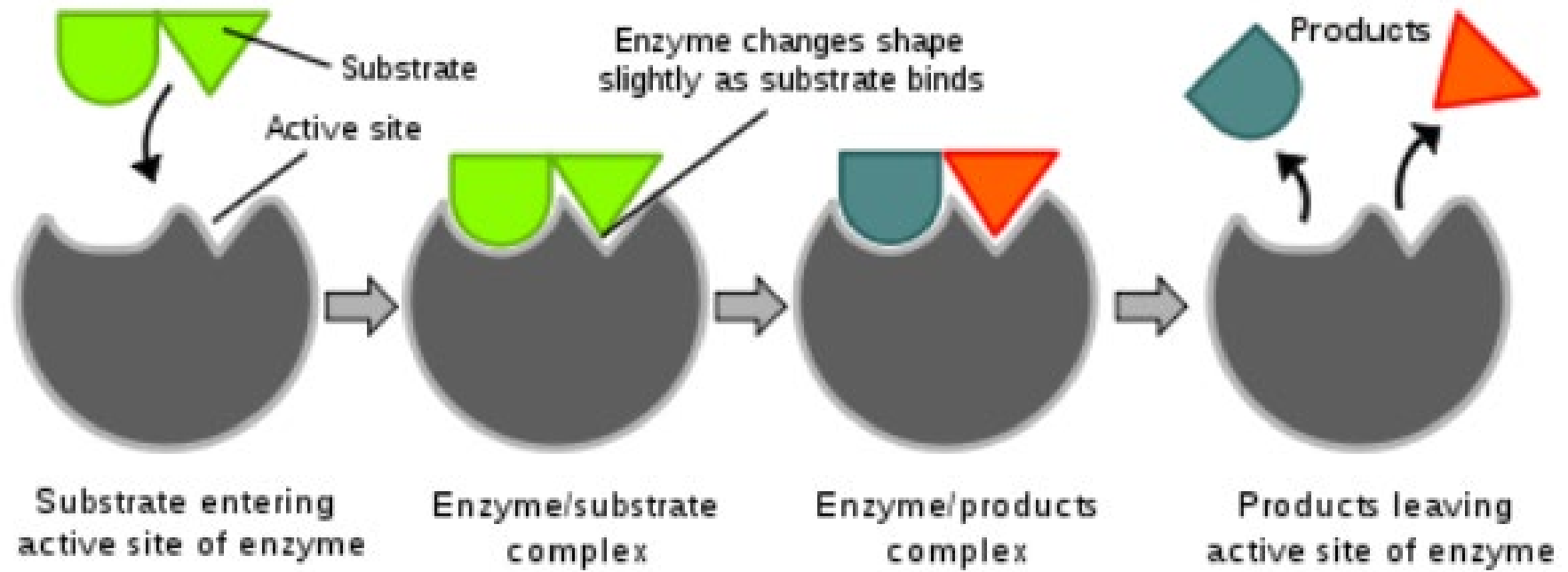


Y-shaped large proteins produced by the immune system to identify foreign entities.

Two top binding identical sites
Key-lock model



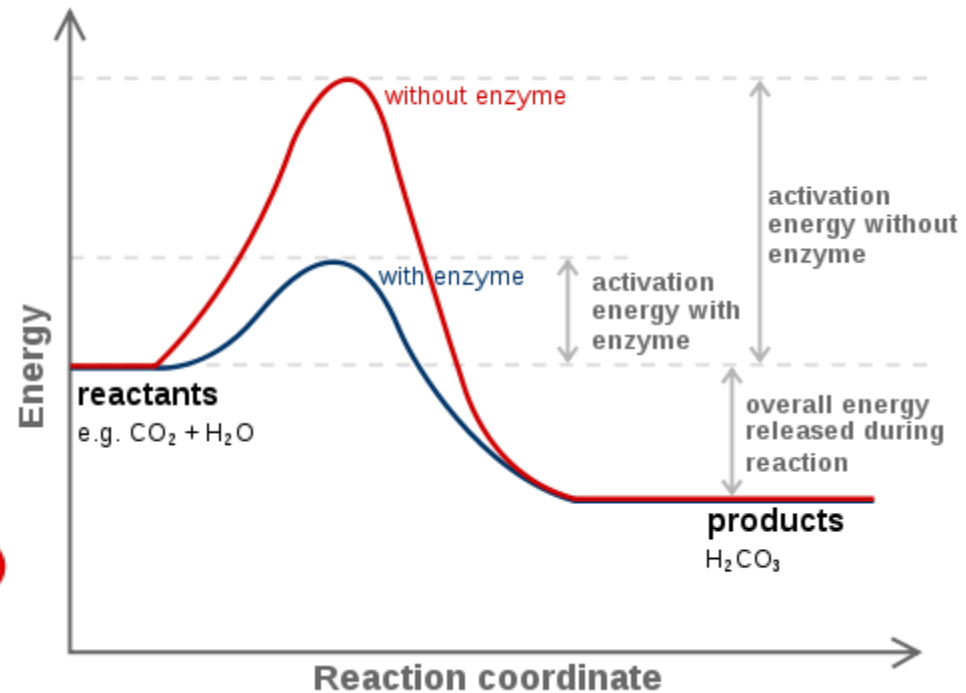
Macromolecules that catalyze specific chemical reactions (metabolic processes): key-lock model



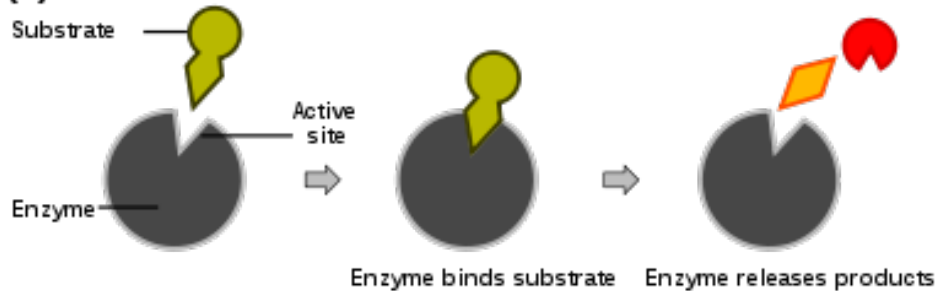


Catalysis and Inhibition

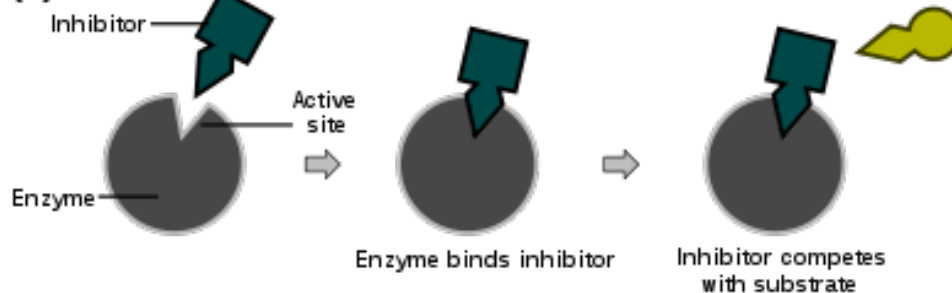
Like any catalyst, enzymes lower the energy barrier



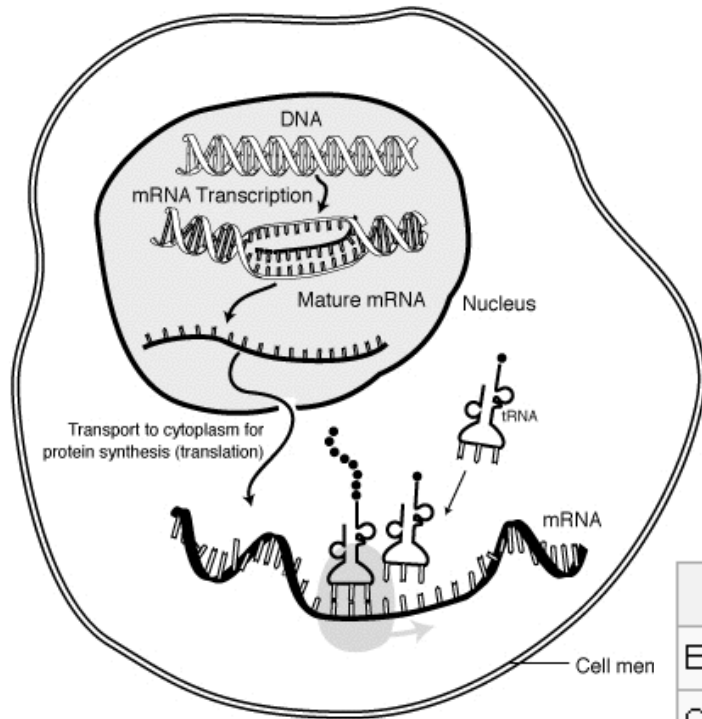
(a) Reaction



(b) Inhibition



Detection can be complicated by inhibitors



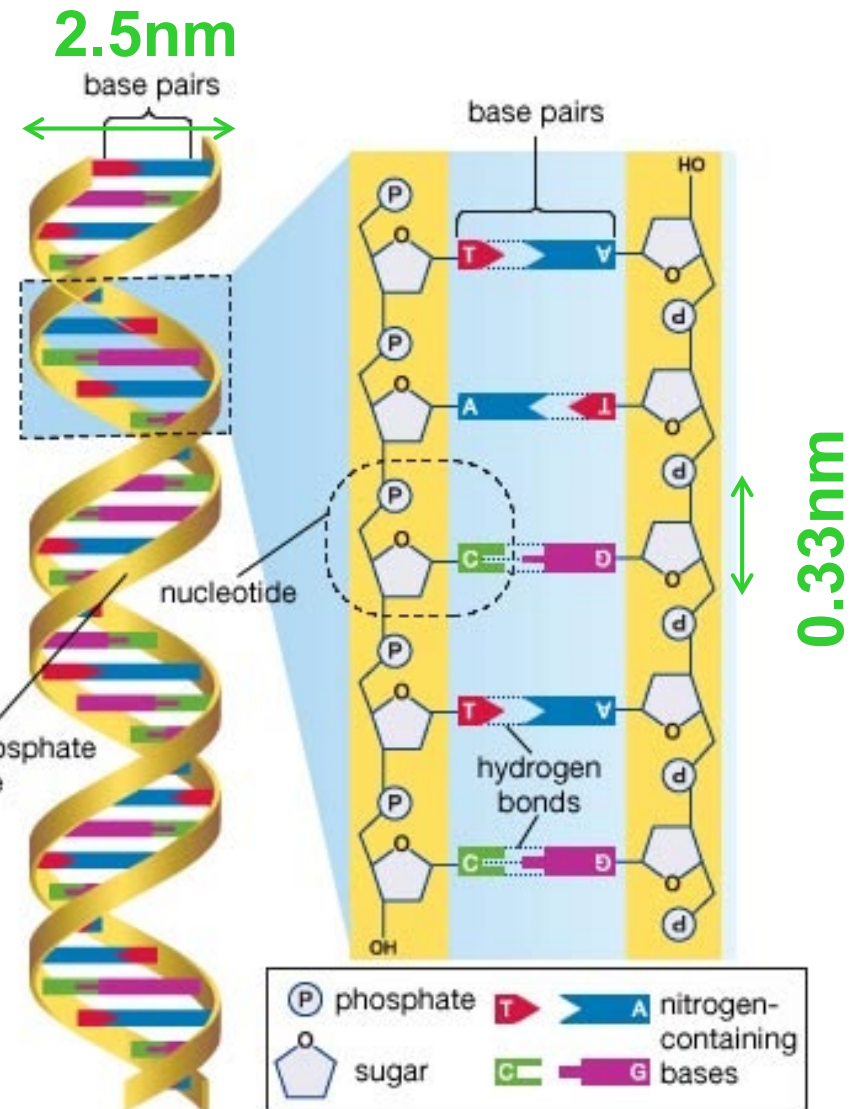
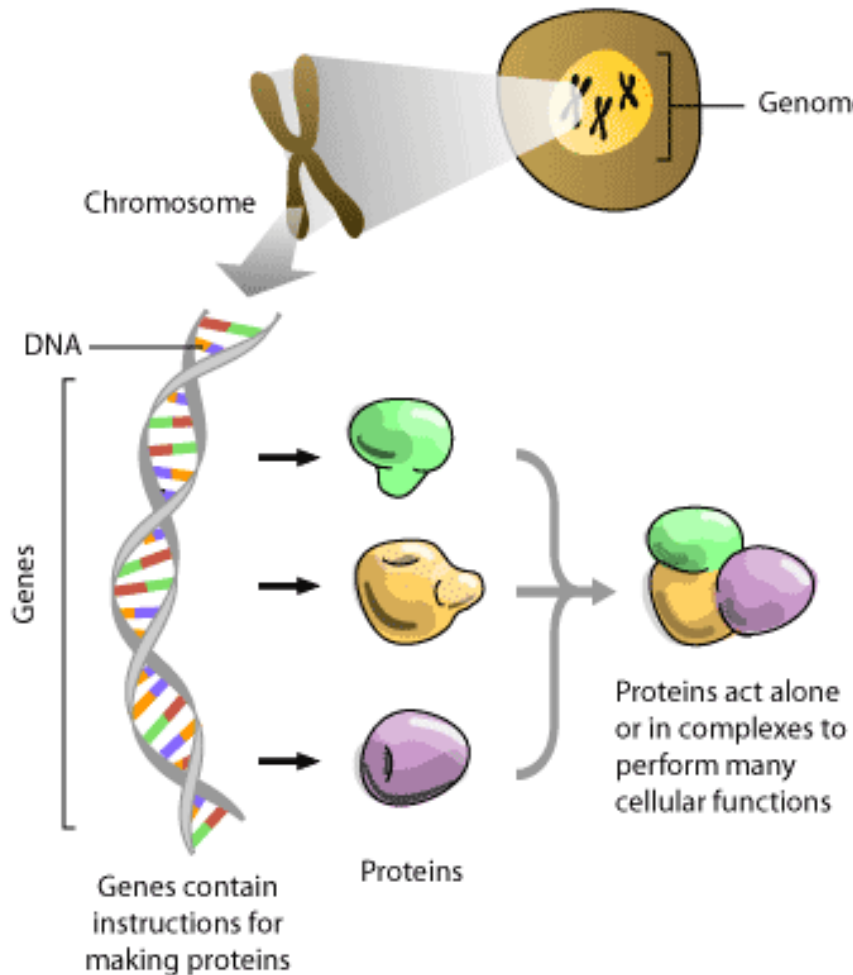
«Central Dogma of Molecular Biology» Flow of genetic information

	DNA	RNA	Proteins
Encodes genetic information	Yes	Yes	No
Catalyzes biological reactions	No	Yes	Yes
Building blocks (type)	Nucleotides	Nucleotides	Amino acids
Building blocks (number)	4	4	20
Strandedness	Double	Single	Single
Structure	Double helix	Highly complex	Highly complex
Stability to degradation	Extremely high	Variable	Variable
Repair systems	Yes	No	No

The Size of the DNA Molecule

Human genome: 23 chromosomes

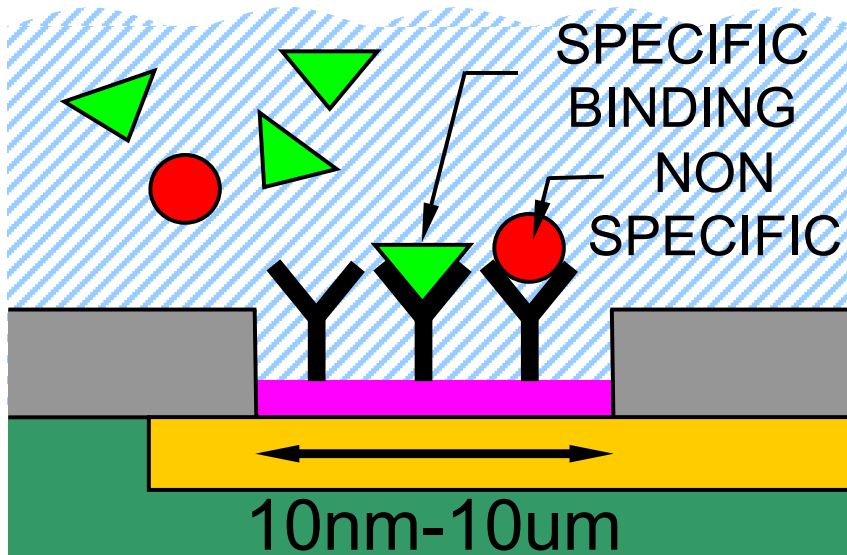
Chr 1 has 250'000kbp





Critical Design Aspects

Molecule selection should be robust to interferences



The specific binding causes a **physical change** (conformational) that can be detected in different ways

Design of the detector **area**:

- sensitivity (signal-to-noise ratio)
- response time (diffusion)
- homogeneity of coverage

Trade-offs analyzed during the course

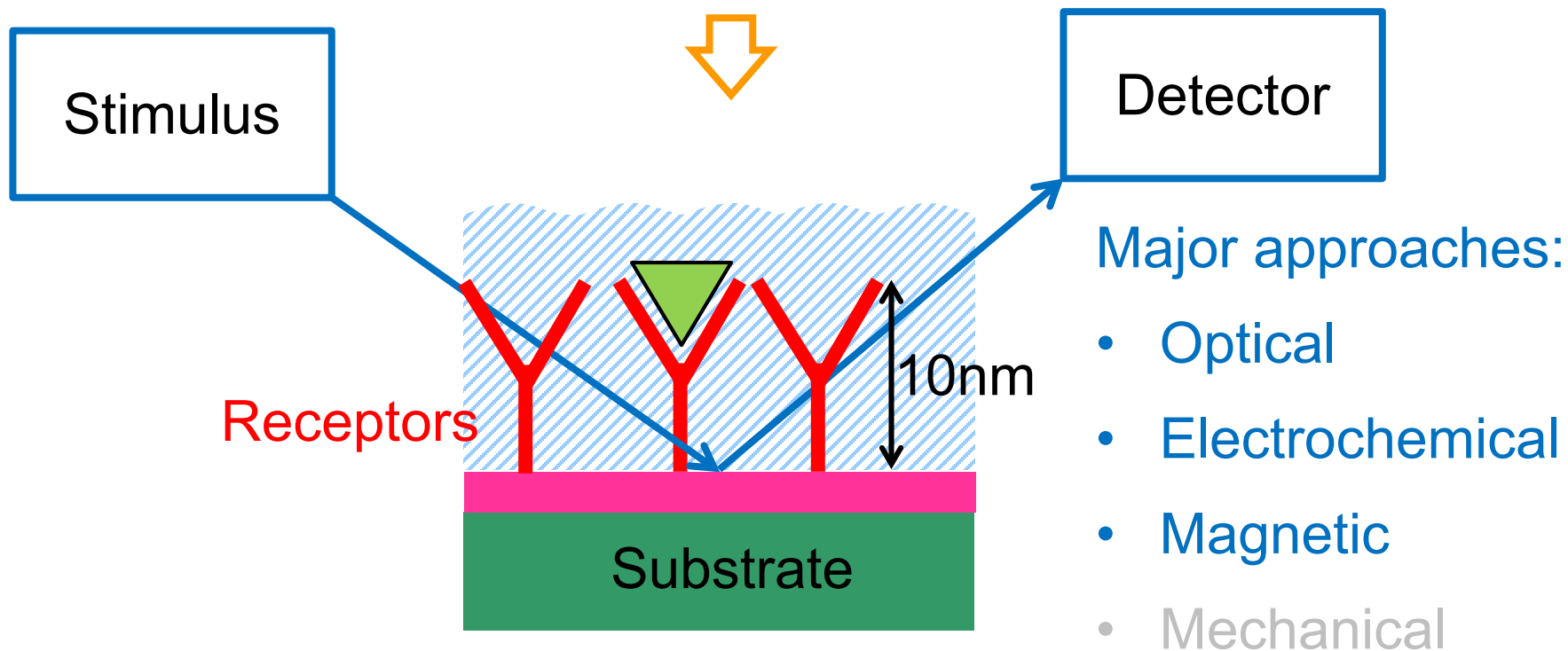


Detection Approaches

Extremely **thin** sensitive layers, in liquid environment



Detection techniques sensitive to planar surface changes

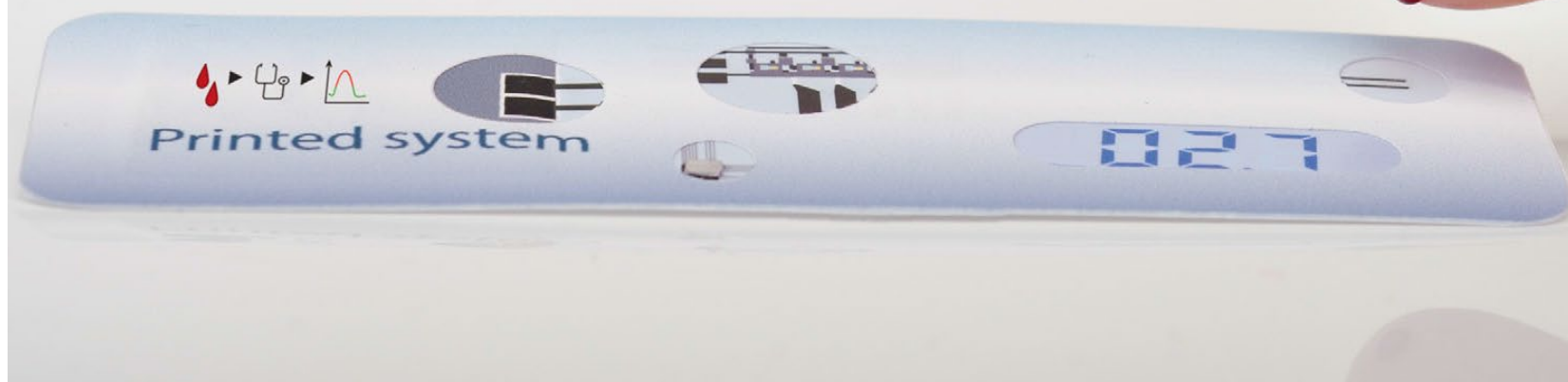


Everything happens at the **interface!**



Introduction to Micro Analytical Systems

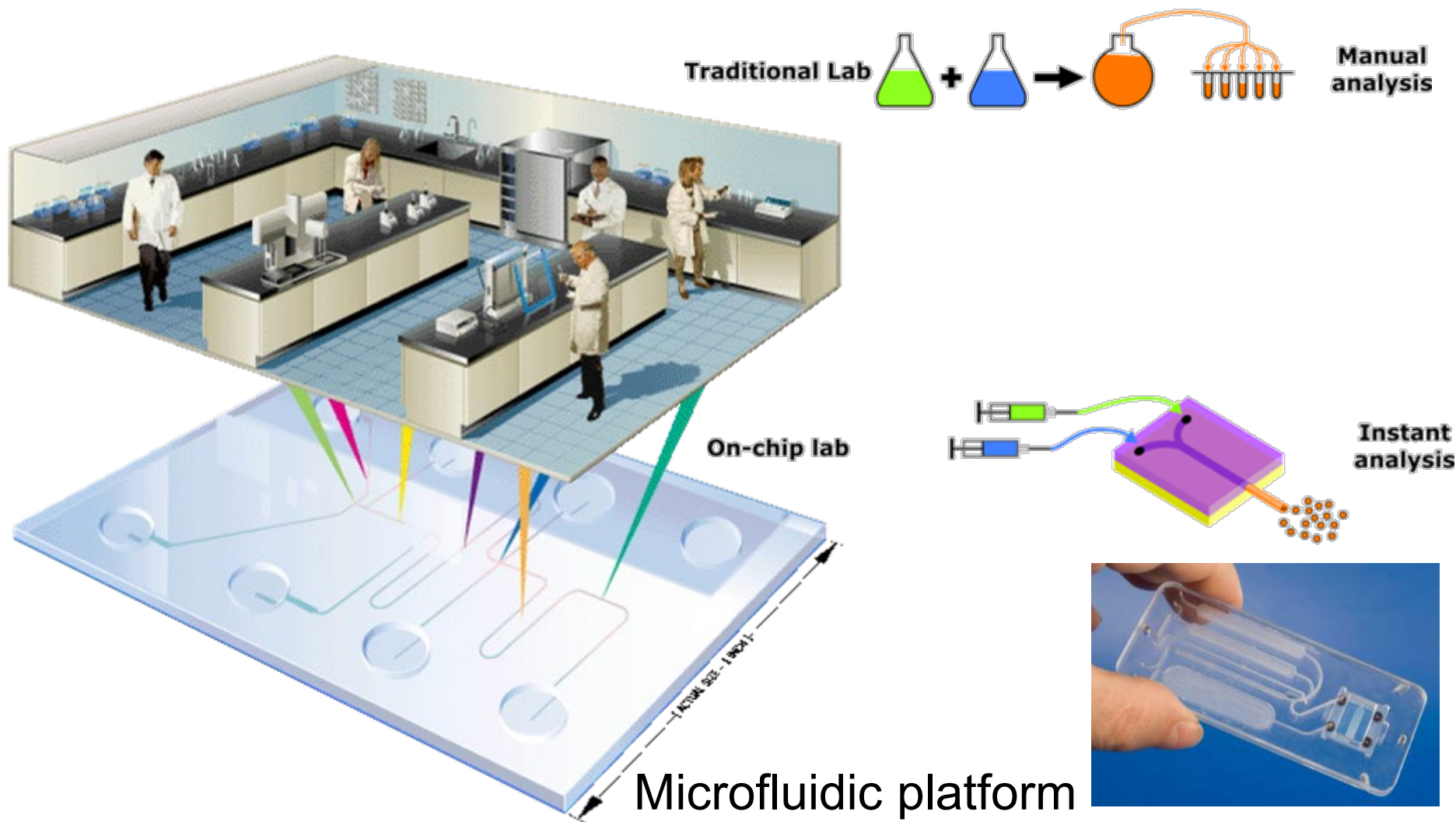
The convergence of material science, micro & nanotechnology, biochemistry and electronics push the development of smart and micro analytical diagnostic chips based on biosensors





The Lab-on-a-Chip Paradigm Shift

Replace lab-scale analysis with integrated miniaturized chips





Advantages of LOC Microsystems

- **Reduced volumes of reagents and samples**, saving money and pollutants

$$10\mu\text{m} \times 10\mu\text{m} \times 10\mu\text{m} = 1\text{pLiter}$$

$$1 \text{ molecule in } 1\text{pLiter} = 1.6\text{pM concentration}$$

- **Better control of fluid dynamics**, non turbulent flows
- **Higher sensitivity** depends on the sensor size
- **Faster response**: at the microscale faster time constants
- **Compact and portable**
- **Ease of use and automatable**: reduced need of external operator actions
- **Low cost** when mass produced

Primary:

- Diagnostics

Secondary:

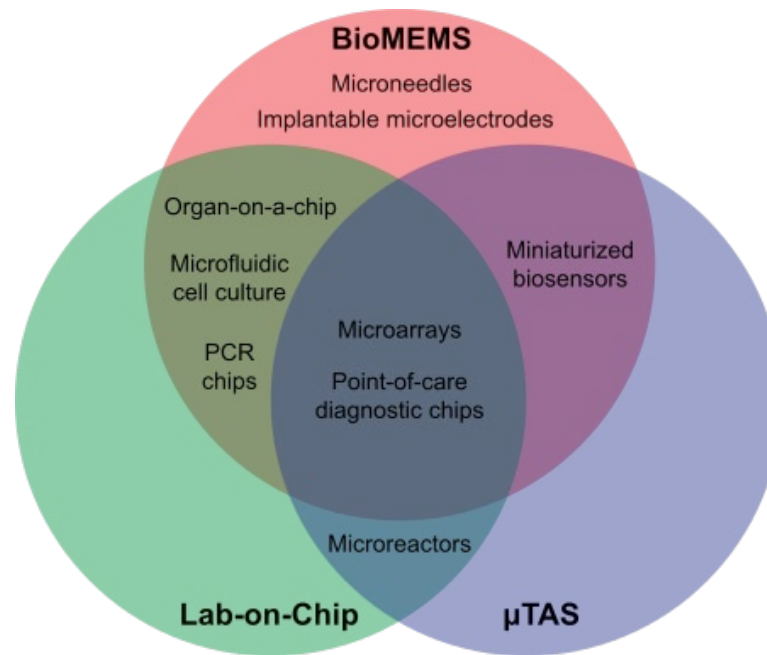
- Agriculture and food quality control
- Industrial processes
- Environmental control (detection of chemical pollutants, water quality)
- Homeland security (detection of toxic agents)





Biochip-Related Keywords and Nomenclature

- *Lab-on-chip* (μ TAS): miniaturized analytical platforms
- *Microfluidics*: the science of miniaturized flow handling
- *BioMEMS*: micromachined systems for bio-chemistry
- *Point of care*: diagnostics at the patient location
- *Soft lithography*: plastic-based fabrication technology





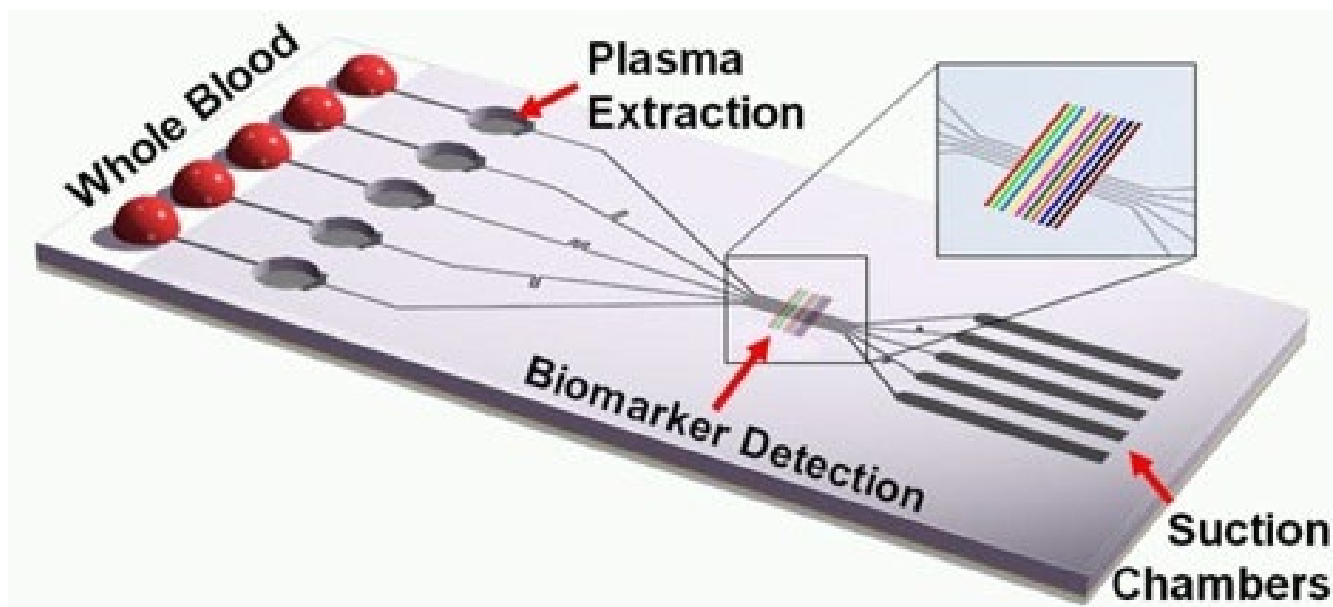
General Architecture: Analysis Flow

Sample
loading

Sample
preparation

Detection

Waste
collection





Course Overview

1. Moving fluids
 2. Moving particles
 3. Microfluidic components
 4. Microfabrication
 5. Optical detection
 6. Magnetic actuation and detection
 7. Electrochemical detection:
 - The electrochemical interface
 - Electrochemical instrumentation
 - Micro and nanoscale electrochemistry
 8. Low-noise current detection
 9. Impedance sensors
- Actuation
- Sensing



Examples of applications

10. Cellular and molecular biosensors

11. DNA

- Introduction and microarrays
- Electrophoresis and thermal issues
- Polymerase Chain Reaction
- Sequencing

12. Design case study: [impedance flow cytometer](#)

13. CMOS Biochips

- State of the art and performances
- Choice of the right substrate (silicon vs. glass)
- Packaging issues



Components and Functions

Functions:

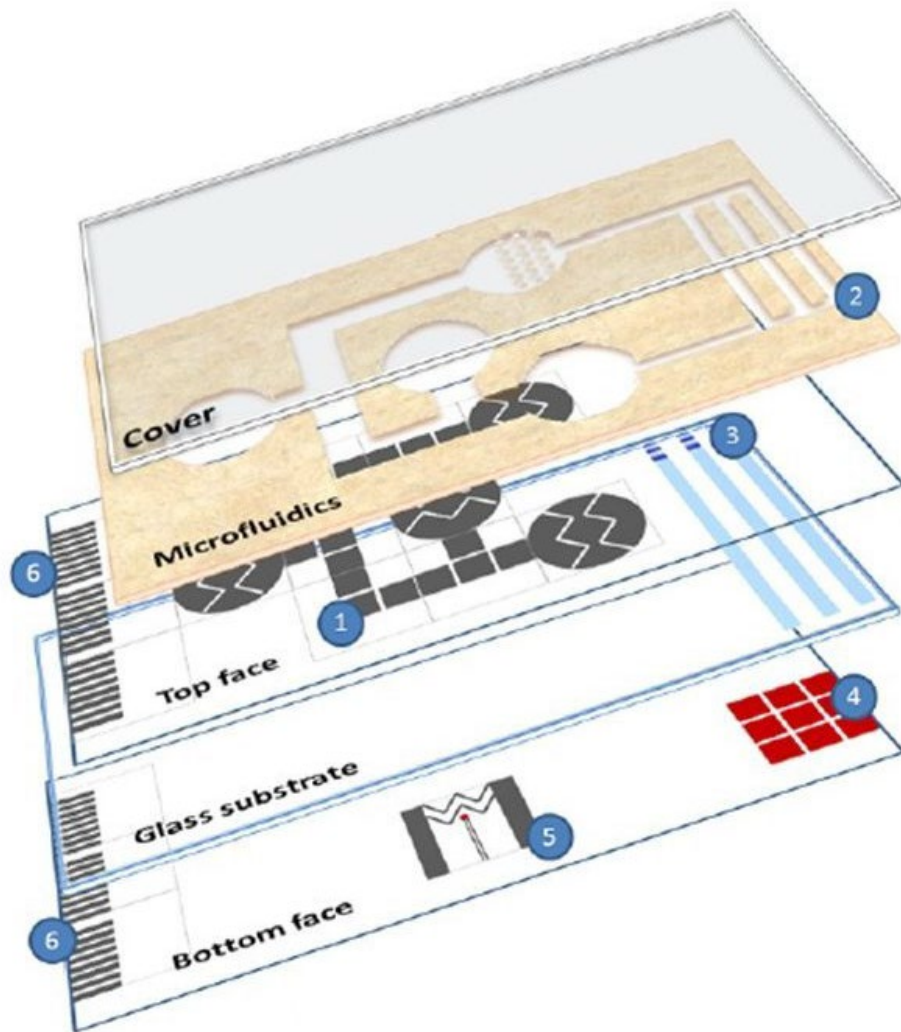
- Sample collection
- Fluid handling
- Sample preparation
(purification, extraction
mixing with reagents, thermal
treatment)
- Detection
- Visualization

Components:

- Inlets - fluidic connectors
- Reagents reservoirs
- Channels and valves
- Pumps
- Mixers
- Heaters
- Biosensors and detectors
- Analog front-end
- Digital elaboration system
- Display & Communication
- Power management



Heterogeneous Materials



LOCs are complex systems usually composed of different materials:

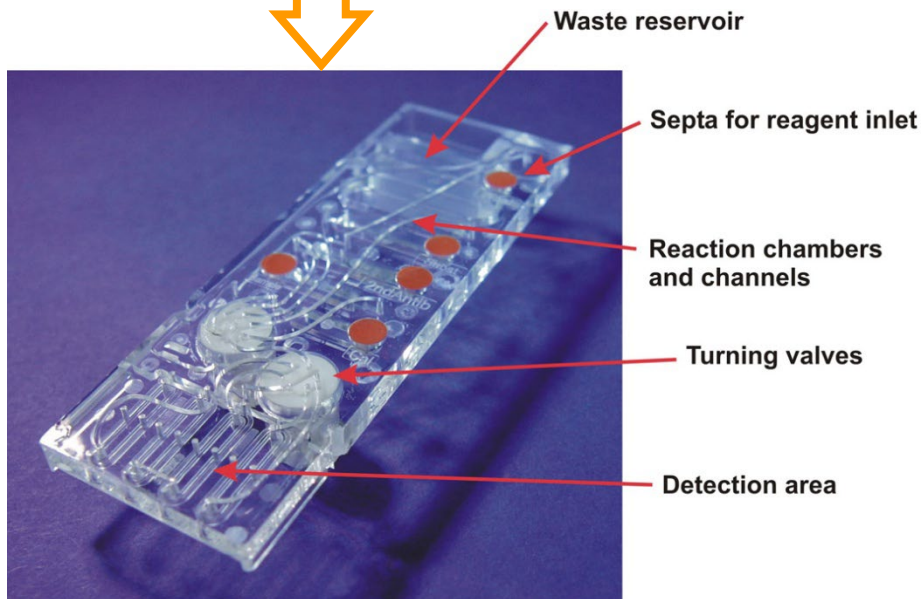
- Glass
- Silicon
- Plastics
- Metals

Different fabrication technologies with different accuracies



Multiple Scenarios of LOC Operation

- **Active** vs passive
- **In-vitro** vs implantable (mainly sensors)
- **Disposable** vs reusable
- **Stand alone** vs **cartridge + reading unit**



- Trained (ambulatory) vs **generic** (home) user

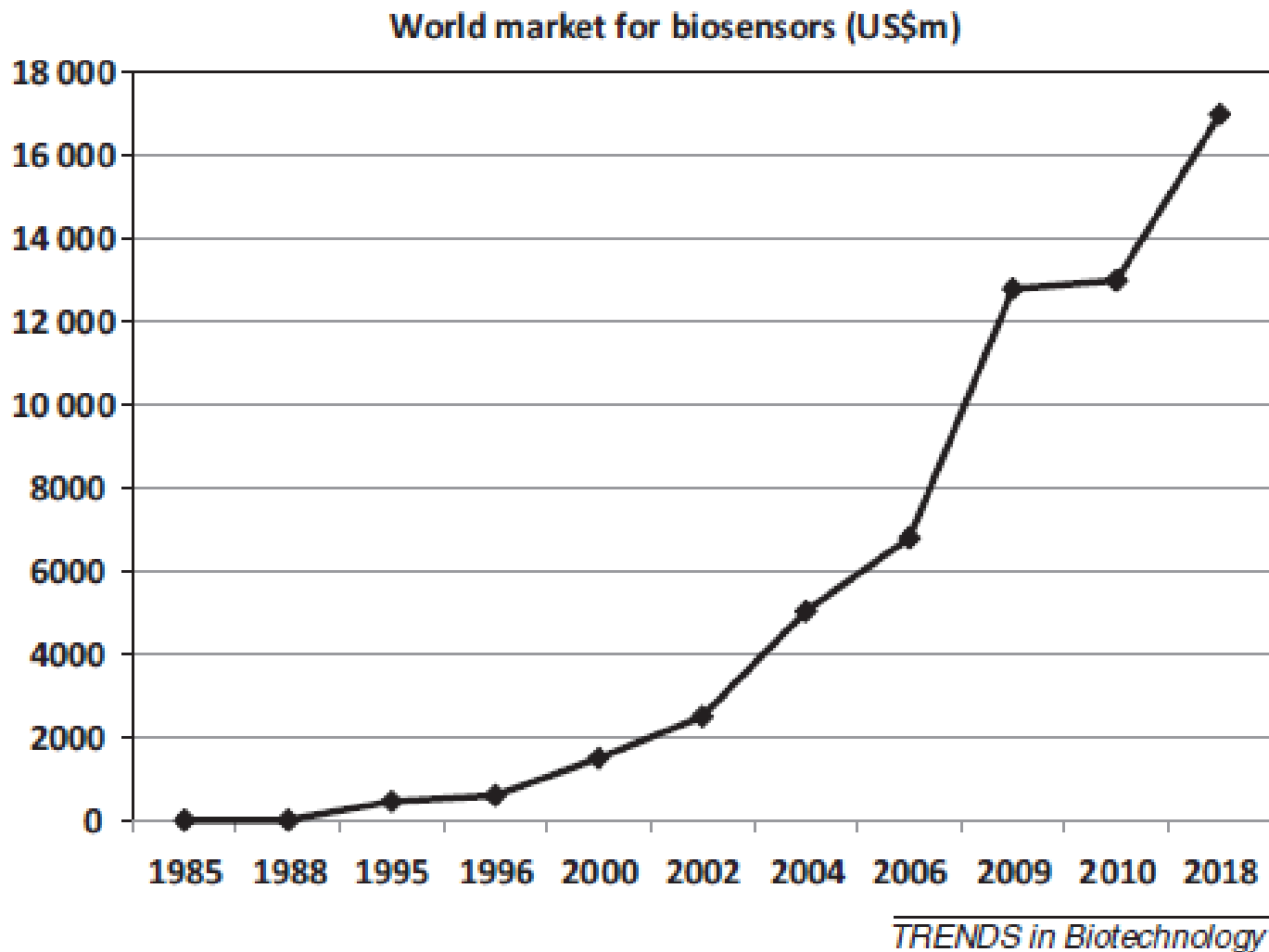


The widespread diffusion of LOC systems for point-of care monitoring is motivated by:

- Earlier diagnosis
- Non invasive detection (high sensitivity required)
- Personalized medicine (continuous monitoring of indicators and assumption of calibrated drugs in **closed-loop**)
- Shift from hospital-based to **home-based assistance**, mainly for control of costs, both in:
 - developed countries: older population increase
 - developing countries: bring access to diagnostics in local and rural communities

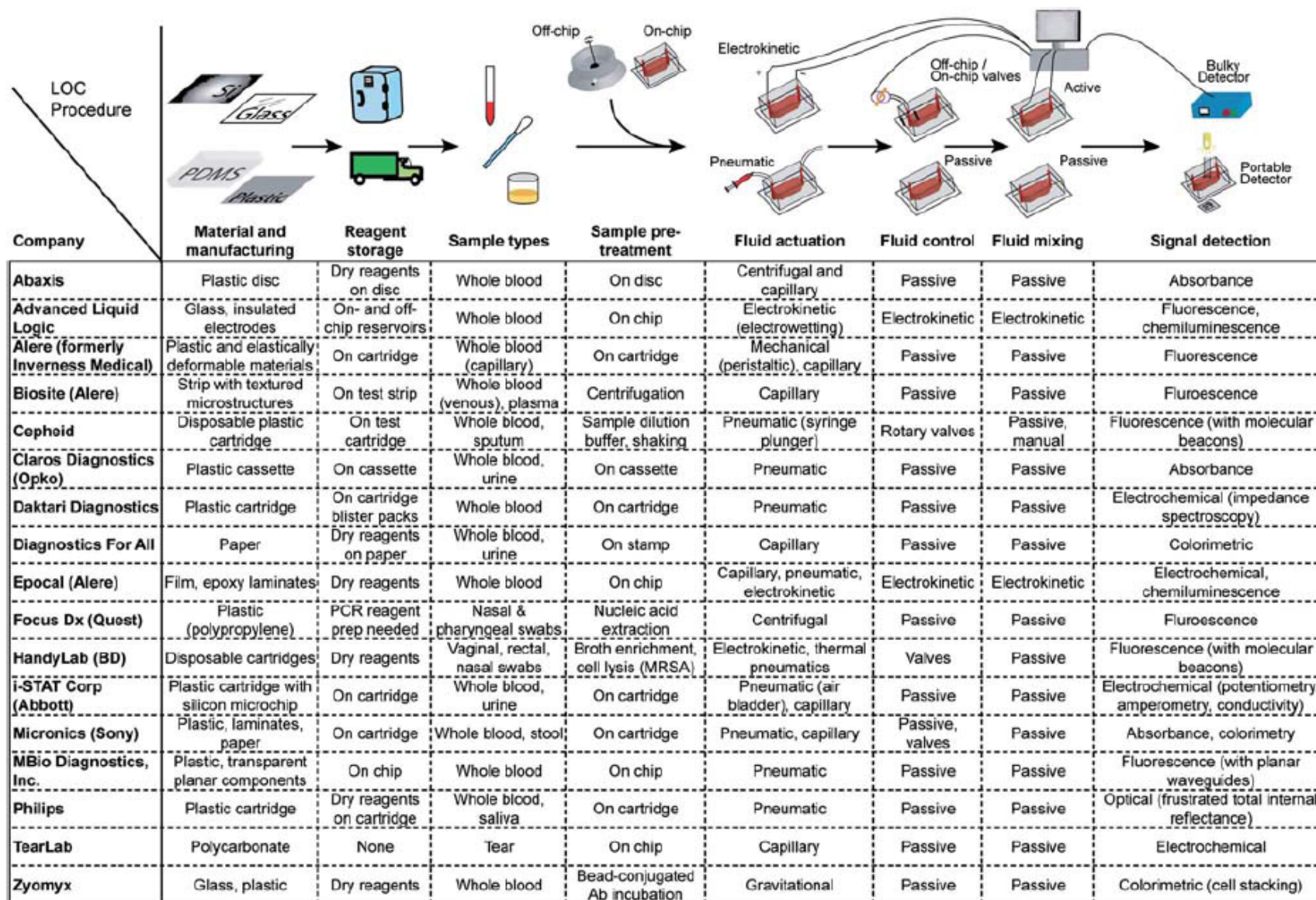


Estimated potential market is huge:





... but so far only a few companies





A Start-Up Example



Daktari
Diagnostics

Microfluidics – Daktari's sample preparation technology, known to specialists as *microfluidic cell chromatography*, isolates cells and other particles in a miniature sensing chamber, without pipetting, labels, or reagents of any kind — none of the complex manual steps that take blood tests out of the hands of clinicians who work beyond the reach of sophisticated labs.



2007 Paper
2009 Product
2014 development
2015 Last funding round

Electrochemical Sensing –

Daktari CD4 also takes advantage of a second innovation, *lysate impedance spectroscopy*. The system uses a simple sensor that counts the captured CD4 cells by measuring their internal contents electrically. A handheld instrument interprets the electrical signal and reports the CD4 count within minutes.

Unencumbered by the lenses, cameras, filters or complex optics that put many diagnostic tests out of reach for clinics, health posts, and mobile caregivers, Daktari CD4 puts the power of accurate test results in the hands of the people who need them.



Two Successful Product Examples: 1. Glucose

Most commercially successful LOCs: *blood glucose monitors*



Enzyme-mediated glucose detection

The enzymatic activity is **electrochemically monitored** since a product of catalysis can be redox detected

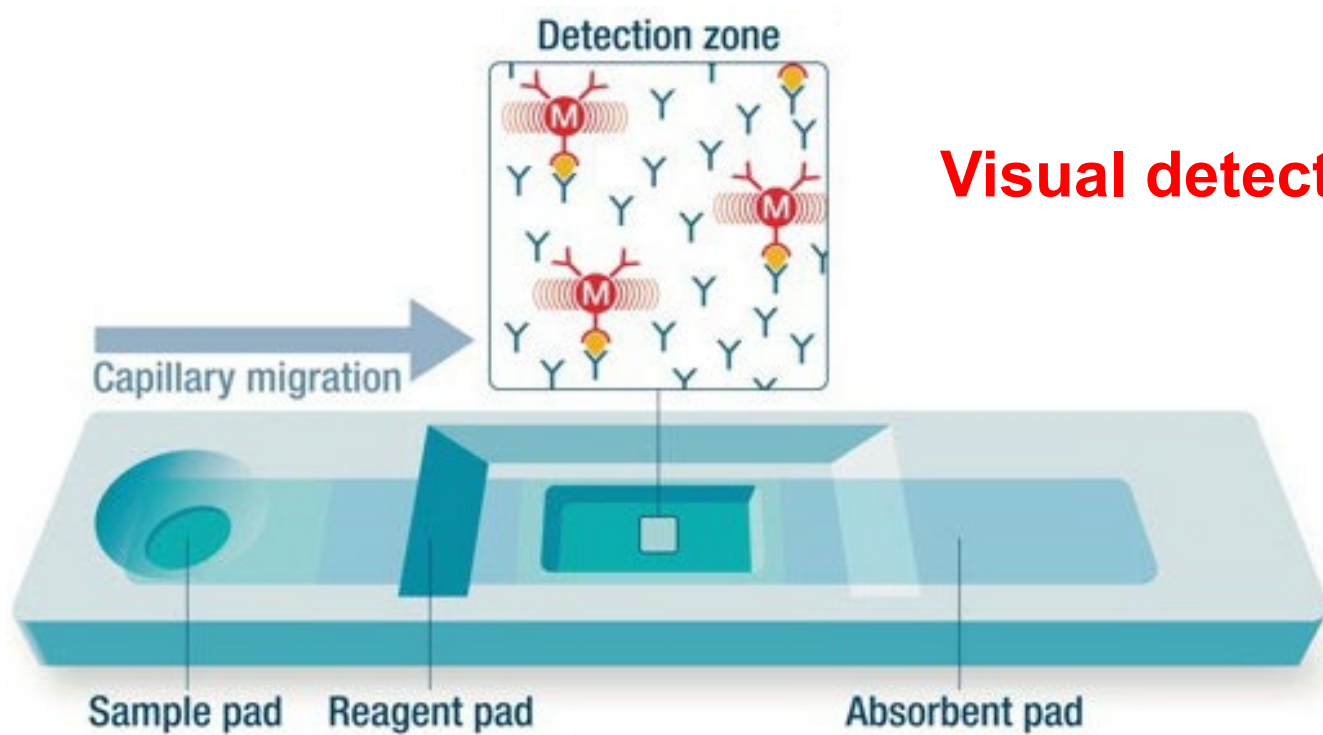
For diabetes, analysis repeated very often

Disposable electrodes prefunctionalized with dried stable enzymes



2. Lateral Flow Test

Passive LOC, loaded with drop of samples moved by capillarity in a porous matrix. Affinity receptors are preloaded in the device.

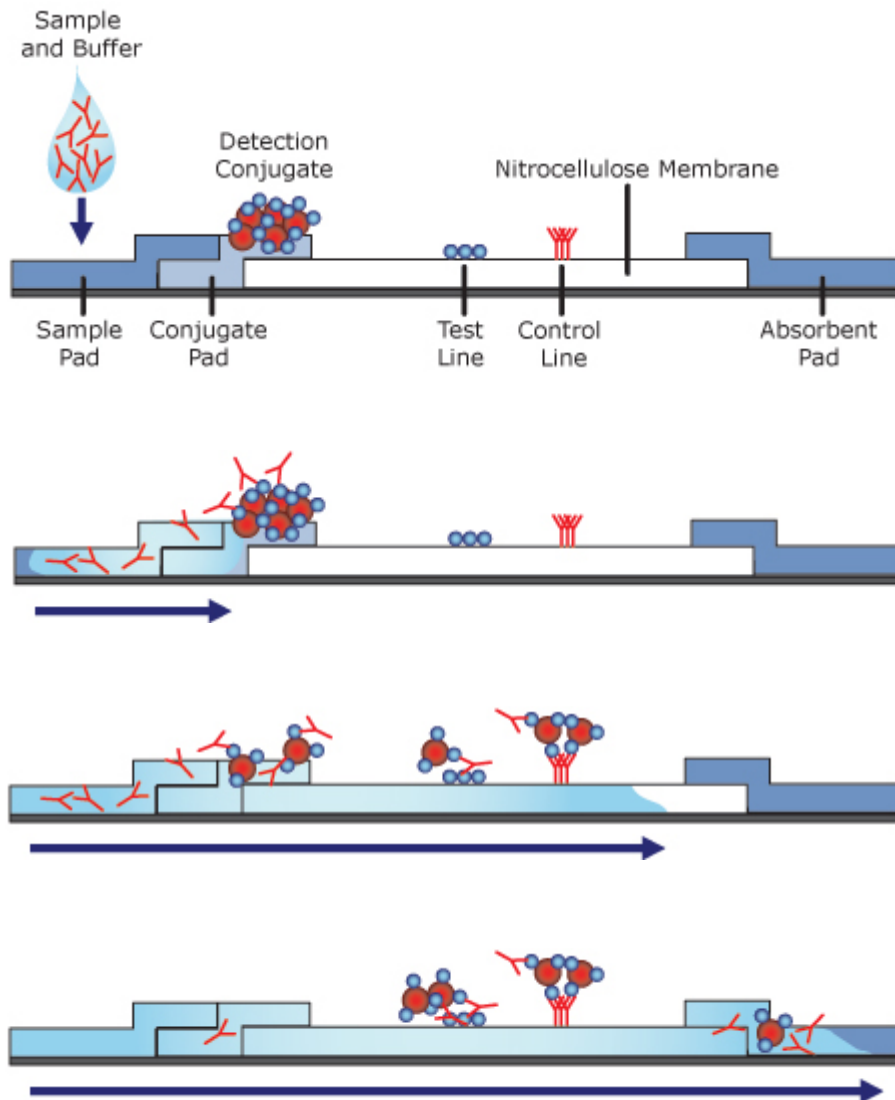


Most common: markers of pregnancy





Working Principle



Loading a sample droplet

dissolves the **receptors** that
are bound to colored
nanoparticles

detection area with fixed
capture receptors

No sample extraction is
performed

How the rapid tests work

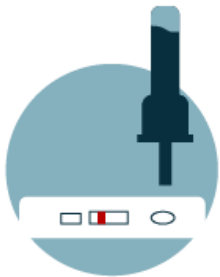
1 Swab from nose or throat



2 Specimen added to extraction tube



3 Drops of extraction solution added to test cartridge



4 Wait 20-30 mins



5 Read results



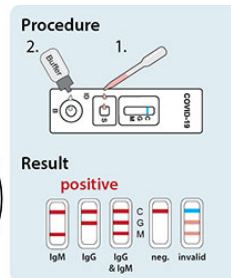
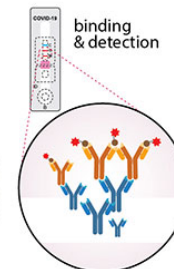
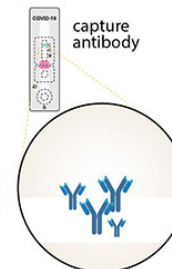
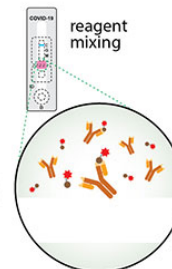
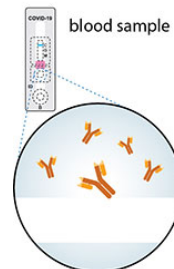
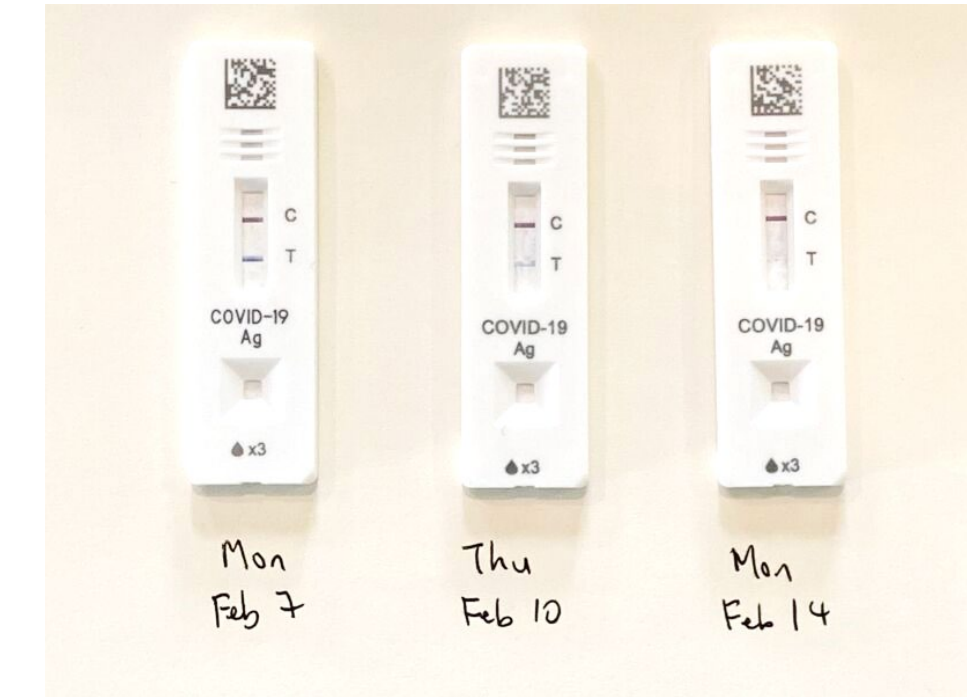
Negative



Positive



Invalid





Key Take-Home Messages

- Biosensors profit from Nature's optimized *molecular affinity* (*enzymes, antibodies, nucleic acids*) for highly selective biochemical identification.
- There is an increasing size compatibility between biological entities and micro/nanotechnology fabrication capability.
- Multidisciplinary convergence for the realization of device and transfer of biochemical protocols from lab-scale to microfluidic platforms with diagnostic and monitoring applications.
- Design solutions, components and technological challenges will be analyzed during this course.
- Huge market potential and social impact (so far only two mass products), but still a lot of work to do! Opportunities for electronic and bio engineers!



Chem Soc Rev

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Biosensors: sense and sensibility†

Cite this: *Chem. Soc. Rev.*, 2013,
42, 3184

Anthony P. F. Turner*

Cite this: DOI: 10.1039/c2lc21204h

www.rsc.org/loc

CRITICAL REVIEW

Commercialization of microfluidic point-of-care diagnostic devices†

Curtis D. Chin,^a Vincent Linder^{*b} and Samuel K. Sia^{*a}

Received 5th December 2011, Accepted 25th January 2012

DOI: 10.1039/c2lc21204h



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